

Lab Report: Deploying NGINX Using Different Base Images and Comparing Image Layers.

Lab Objectives:

- Deploy NGINX using Official, Ubuntu-based, and Alpine-based images.
 - Understand Docker image layers and size differences.
 - Compare performance, security, and use-cases of each approach.
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Part 1: Deploy NGINX Using Official Image (Recommended Approach)

The official NGINX image is pre-optimized, uses a Debian-based OS internally, and requires minimal configuration.

Step 1: Pull the Image

We retrieved the latest official NGINX image from the Docker registry.

Command: `docker pull nginx:latest`

Step 2: Run the Container

We ran the container in detached mode (-d) naming it nginx-official and mapped port 8080 on the host to port 80 in the container.

Command: `docker run -d --name nginx-official -p 8080:80 nginx`

Step 3: Verify

We verified the deployment by accessing the localhost URL. The response confirms the NGINX welcome page is active.

Command: `curl http://localhost:8080`

Key Observations

We checked the size of the downloaded official image.

Observation: The official image size is approximately 161MB.

Part 2: Custom NGINX Using Ubuntu Base Image

In this section, we built a custom image using ubuntu:22.04 as the base. This approach results in a larger image but provides full OS utilities, which is useful for debugging.

Step 1: Build the Image

We created a Dockerfile that installs NGINX on top of Ubuntu and built the image with the tag nginx-ubuntu.

Command: docker build -t nginx-ubuntu .

Step 2: Run the Container

We ran the Ubuntu-based container on port 8081.

Command: docker run -d --name nginx-ubuntu -p 8081:80 nginx-ubuntu

Observations

The resulting image is significantly larger due to the inclusion of full OS utilities.

Observation: The image size is approximately 134MB.

Part 3: Custom NGINX Using Alpine Base Image

We built a custom image using alpine:latest. Alpine Linux is a security-oriented, lightweight Linux distribution.

Step 1: Build Image

We built the image tagged nginx-alpine.

Command: docker build -t nginx-alpine .

Step 2: Run the Container

We ran the Alpine-based container mapping port 8082 to port 80.

Command: docker run -d --name nginx-alpine -p 8082:80 nginx-alpine

Observations

The Alpine-based image is extremely small compared to the others.

Observation: The image size is approximately 10.4MB.

Part 4: Image Size and Layer Comparison

Compare Sizes

We compared all three images side-by-side to visualize the size differences.

Inspect Layers

We inspected the layer history of all three images.

- **Ubuntu** has many filesystem layers due to the full OS utilities.
- **Alpine** has minimal layers, contributing to its small size.
- **Official NGINX** is optimized but heavier than Alpine.

Command: docker history nginx, docker history nginx-ubuntu, docker history nginx-alpine.

Part 5: Functional Tasks (Serving Custom Content)

Step 1: Create Custom HTML

We created a directory named html and added a custom index.html file.

Commands:

- mkdir html
- echo "<h1>Hello from Docker NGINX</h1>" > html/index.html

Step 2: Run with Volume Mount

We ran a new container, mounting the local html directory to /usr/share/nginx/html inside the container. This allows NGINX to serve our custom file.

Command: docker run -d -p 8083:80 -v \$(pwd)/html:/usr/share/nginx/html nginx

Part 6: Comparison Summary

Feature	Official NGINX	Ubuntu NGINX	Alpine NGINX
Image Size	Medium	Large	Very Small
Ease of Use	Very Easy	Medium	Medium
Startup Time	Fast	Slow	Very Fast
Debugging Tools	Limited	Excellent	Minimal
Security Surface	Medium	Large	Small
Production Ready	Yes	Rarely	Yes

Part 7: When to Use What

- **Official NGINX Image:** Recommended for production deployment, standard web hosting, and reverse proxy/load balancer use cases.
- **Ubuntu-Based Image:** Best for learning Linux/NGINX internals or when heavy debugging and custom system-level dependencies are required.
- **Alpine-Based Image:** Ideal for microservices, CI/CD pipelines, and cloud/Kubernetes workloads due to its small footprint.